IEDA 4000H

Tutorial 4

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- $X = X^{\frac{1}{2}} X^{\frac{1}{2}}$ (how? eigen-decomposition)
- $\det(X + tV) = \det(X)\det(I + tX^{-\frac{1}{2}} V X^{-\frac{1}{2}})$ $\big($why? $\det(AB) = \det(BA) = \det(A)\det(B)\big)$
- Let $\tilde{V} = X^{-\frac{1}{2}} V X^{-\frac{1}{2}}$. Then $I + t\tilde{V} \succ 0$. (why?)
- Let μ_i be the i-th eigenvalue of $\tilde{V}$. What are the eigenvalues of $I + t\tilde{V}$?

Exercise

Find the Lagrangian dual of the following constrained quadratic programming problem:

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & \frac{1}{2}\boldsymbol{x}^\top Q\boldsymbol{x} + \boldsymbol{c}^\top \boldsymbol{x} \\ \text{s.t.} \quad & A\boldsymbol{x} = \boldsymbol{b}, \\ & \boldsymbol{x} \geq 0, \end{aligned}$$

where $Q \in \mathbb{R}^{n\times n}$ is positive definite, $A \in \mathbb{R}^{m\times n}$, $\boldsymbol{b} \in \mathbb{R}^m$ and $\boldsymbol{c} \in \mathbb{R}^n$.

Solution:

The Lagrangian function is

$$L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = \frac{1}{2}\boldsymbol{x}^\top \mathrm{Q}\boldsymbol{x} + \boldsymbol{c}^\top \boldsymbol{x} - \boldsymbol{\lambda}^\top (\mathrm{A}\boldsymbol{x} - \boldsymbol{b}) - \boldsymbol{\mu}^\top \boldsymbol{x},$$

where $\boldsymbol{\lambda} \in \mathbb{R}^m$ and $\boldsymbol{\mu} \geq 0 \in \mathbb{R}^n$ are the Lagrange multipliers associated with the equality and inequality constraints, respectively. The Lagrangian dual function is given by

$$h(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \inf_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = -\frac{1}{2}(\mathrm{A}^\top \boldsymbol{\lambda} + \boldsymbol{\mu} - \boldsymbol{c})^\top \mathrm{Q}^{-1}(\mathrm{A}^\top \boldsymbol{\lambda} + \boldsymbol{\mu} - \boldsymbol{c}) + \boldsymbol{\lambda}^\top \boldsymbol{b},$$

for $\mathrm{A}^\top \boldsymbol{\lambda} + \boldsymbol{\mu} - \boldsymbol{c} \in \mathrm{range}(\mathrm{Q})$ (when Q is positive semi-definite); otherwise, $h(\boldsymbol{\lambda}, \boldsymbol{\mu}) = -\infty$.

Let $\boldsymbol{y} = A^\top \boldsymbol{\lambda} + \boldsymbol{\mu} - \boldsymbol{c}$. The Lagrangian dual problem is

$$\begin{aligned} \max_{\boldsymbol{y},\, \boldsymbol{\lambda}} \quad & -\frac{1}{2}\boldsymbol{y}^\top Q^{-1}\boldsymbol{y} + \boldsymbol{b}^\top \boldsymbol{\lambda} \\ \text{s.t.} \quad & \boldsymbol{y} + \boldsymbol{c} - A^\top \boldsymbol{\lambda} \geq 0. \end{aligned}$$

Duality Gap

Consider the following constrained optimization problem:

$$
\begin{aligned}
\min_{x,\, y} \quad & e^{-x} \\
\text{s.t.} \quad & \frac{x^2}{y} \le 0, \\
& y > 0.
\end{aligned}
$$

1. Find the optimal value of the primal problem.
2. Find the Lagrangian dual of the primal problem and its optimal value.
3. What is the duality gap?

Solution:

The optimal value of the primal problem is 1, which is attained at $(x, y) = (0, y^*)$, $y^* > 0$. The Lagrangian function is

$$L(x, y, \lambda) = e^{-x} + \lambda \frac{x^2}{y},$$

where $\lambda \geq 0$. The Lagrangian dual function is given by

$$h(\lambda) = \inf_{x \in \mathbb{R},\ y > 0} L(x, y, \lambda) = \inf_{x \in \mathbb{R},\ y > 0} \left(e^{-x} + \lambda \frac{x^2}{y} \right) = 0.$$

(Let $x \to \infty$ and $y = x^4 \to \infty$.)

The Lagrangian dual of the primal problem is

$$\begin{aligned} \max_{\lambda} \quad & 0 \\ \text{s.t.} \quad & \lambda \geq 0. \end{aligned}$$

The optimal value of the dual problem is obviously 0. Thus, the duality gap is 1.

Exercise

Consider the following optimization problem:

$$\begin{aligned} \min_{x,\, y} \quad & x^2 + y^2 \\ \text{s.t.} \quad & y^2 \leq x - 1, \\ & x \leq 2. \end{aligned}$$

Find the KKT conditions and the optimal solution.

Solution:

The Lagrangian function is

$$L(x, y, \lambda, \mu) = x^2 + y^2 + \lambda(y^2 - x + 1) + \mu(x - 2),$$

where $\lambda \geq 0$ and $\mu \geq 0$.

The KKT conditions are

- Stationarity:

$$\frac{\partial L}{\partial x} = 2x - \lambda + \mu = 0, \quad \frac{\partial L}{\partial y} = 2y + 2\lambda y = 0.$$

- Primal feasibility:

$$y^2 - x + 1 \le 0, \quad x - 2 \le 0.$$

- Dual feasibility:

$$\lambda \ge 0, \quad \mu \ge 0.$$

- Complementary slackness:

$$\lambda(y^2 - x + 1) = 0, \quad \mu(x - 2) = 0.$$

- Case 1 ($\lambda = 0, \mu = 0$):
$$2x - \lambda + \mu = 0 \implies x = 0,$$
thus this case is infeasible.
- Case 2 ($\lambda = 0, \mu > 0$):
$$\mu(x-2) = 0 \implies x = 2$$
$$2x - \lambda + \mu = 0 \implies \mu = -4 < 0,$$
thus this case is dual infeasible.

- Case 3 ($\lambda > 0, \mu = 0$):

$$\begin{aligned}
\lambda(y^2 - x + 1) = 0 &\implies y^2 - x + 1 = 0,\\
2y + 2\lambda y = 0 &\implies y = 0,\\
y^2 - x + 1 = 0 &\implies x = 1,\\
2x - \lambda + \mu = 0 &\implies \lambda = 2,
\end{aligned}$$

  thus $(x, y, \lambda, \mu) = (1, 0, 2, 0)$ is a KKT point.

- Case 4 ($\lambda > 0, \mu > 0$):

$$\begin{aligned}
\lambda(y^2 - x + 1) = 0 &\implies y^2 - x + 1 = 0,\\
\mu(x - 2) = 0 &\implies x = 2,\\
2y + 2\lambda y = 0 &\implies y = 0,\\
y^2 - x + 1 = 0 &\implies x = 1,
\end{aligned}$$

which leads to a contradiction.

Therefore, the only KKT point is $(x, y, \lambda, \mu) = (1, 0, 2, 0)$, which is also the optimal solution to the primal problem.

Thank you for listening !

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